

Active Management of IR Exposure

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IR Exposure Allocation Strategy: June 2016 Outlook

- ▶ **Fixed Income Fund Allocation:** 3.4bps (May: 4.6bps%)
- ▶ **IR Exposure Model:** -1.5bps% (May: NA)

Active Management of IR Exposure

Investigate if we can design an **active** strategy for investing in IR Exposure that outperforms a **passive** strategy that doesn't hedge spread exposure.

Metrics and Set-up

- ▶ Minimize the volatility of Portfolio PnL corresponding to spread and rates changes
 - ▶ Spread PnL: Historical values for Portfolio CMO Spread Sensitivity multiplied by the corresponding spread changes for the benchmark IO.¹
 - ▶ Rates PnL: Historical values for Portfolio Interest Rate Sensitivity multiplied by the corresponding yield changes for the 10-year.

¹The benchmark IO is taken to be IOS 4s of 2010. The spread changes are a combination of OAS changes for the Barclays and YieldBook models.

Framework for IR Exposure Allocation Strategy

- ▶ Minimize Portfolio PnL corresponding to spread and rates changes by dynamically setting IR exposure.
- ▶ Potential variables influencing IR exposure
 - ▶ Correlations between yield changes and IO spread changes: 60-day moving average of correlations between changes in 10-year Treasury yields and changes in benchmark IO OASs.
 - ▶ Convexity: 60-day moving average of realized volatility on the current coupon
 - ▶ Prepayment Risk: Incentive for the Mortgage Universe
 - ▶ IO Exposure: Net IO
 - ▶ IO OAS
 - ▶ Economic variables: GDP NowCast

Active IR Exposure Allocation Strategy: Optimization Results

- ▶ Significant Variables
 - ▶ High: Correlation, Net IO
 - ▶ Medium: Prepayment Risk, Convexity
 - ▶ Low: GDP NowCast
- ▶ Final Model
 - ▶ Proj. IR Exposure $\sim$ Correlation + Net IO